# Lecture 12

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# Why Evaluate Portfolio Performance?

- The majority of financial assets are managed by professional investors
  - Mutual funds
  - Pension funds
  - Endowments
  - Hedge funds
  - Insurance portfolios
- Capital owners delegate investment decisions → agency problem arises
- Evaluation serves:
  - Manager selection
  - Compensation and incentives
  - Capital allocation across strategies
  - Determination of passive vs active implementation

# The Core Question: Disentangling Return from Performance

- Observed returns alone do not measure skill
  - High returns may reflect high risk
  - Low returns may reflect conservative positioning
- Central challenge is to distinguish absolute returns from performance
  - How much of a portfolio's return reflects compensation for risk, and how much reflects abnormal performance?
  - CAPM framework: $R_p = R_f + \beta_p(R_M - R_f) + \alpha_p$
- Total return =
  - Risk-free compensation
  - Systematic risk exposure
  - Residual abnormal return
- Performance evaluation is inherently model-dependent:
  - $\alpha_p = R_p - E[R_p | \text{risk model}]$
  - Skill can only be measured relative to a benchmark model.

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# Arithmetic vs Geometric: Why We Need Both

- Recollect that:
  - Arithmetic mean: $\bar{R} = \frac{1}{T}\sum_{t=1}^{T} R_t$
  - Geometric mean: $R_g = \left(\prod_{t=1}^{T}(1+R_t)\right)^{1/T} - 1$
  - Relationship (approx.): $R_g \approx \bar{R} - \frac{1}{2}\sigma^2$
- Arithmetic average → forward-looking modeling
  - Expected one-period return
  - Input to asset pricing models
  - Used in mean–variance optimization
  - Relevant for Sharpe ratio and risk models
- Geometric average → realized investor outcome
  - Compound growth rate
  - Measures realized wealth accumulation
  - Directly linked to terminal wealth
- Investors ultimately care about :max $E[W_T]$

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# Money-Weighted (Dollar-Weighted) Return

- Money-weighted return (IRR) solves:

$$0 = \sum_{t=0}^{T} \frac{CF_t}{(1 + IRR)^t}$$

- Incorporates timing and magnitude of cash flows
- Reflects actual investor experience
- Can differ substantially from geometric average
- Difference arises because:
  - Geometric return assumes fixed capital base
  - IRR weights returns by invested dollars
- IRR appropriate when:
  - Capital is committed gradually
    - Private equity
    - Venture capital
    - Real estate funds
      - Manager controls timing of capital deployment
- Not appropriate for evaluating manager skill when:
  - Cash flows are driven by investor decisions
  - Flows are exogenous to manager

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# Adjusting Returns for Risk

- Absolute return ≠ performance.
- Performance must adjust for:
  - The level of risk taken
  - The type of risk taken
  - The investor's portfolio context
- General principle: $\text{Performance} = \frac{\text{Excess Return}}{\text{Relevant Risk}}$
- "Relevant Risk" depends on the decision problem
- Different measures are appropriate for different purposes:
  - Standalone portfolio evaluation
  - Selection of components for a diversified risky portfolio
  - Active management relative to benchmark
  - Evaluation of abnormal performance relative to pricing model
- Risk adjustment is not one-size-fits-all.

# The Comparison Universe

- Performance is relative to a benchmark or universe.
- A simple way to look at this is through a “Comparison Universe”
  - Examples:
    - Broad market benchmark: S&P 500, MSCI World
    - Asset class benchmark: Russell 2000, Bloomberg Aggregate Bond Index
    - Peer universe: Large-cap growth funds, Emerging market equity funds
    - Liability benchmark: Pension funding target
  - The benchmark serves as a proxy measure for:
    - Expected return
    - Relevant risk
    - Opportunity cost

# Sharpe Ratio and $M^2$

- Sharpe ratio:
  - Uses total volatility
  - Measures excess return per unit of total risk
  - Equal to slope of the Capital Market Line
  - Appropriate when:
    - Portfolio is entire risky allocation
    - Investor holds only this risky portfolio + risk-free asset
- $M^2$ measure: $M^2=R_f+S_p\sigma_M$
  - Converts Sharpe into percentage return
  - Interpretable in return units
  - Ranking identical to Sharpe

# Treynor Measure

- Treynor ratio: $T_p = \frac{E(R_p - R_f)}{\beta_p}$
- Uses systematic risk only.
- Rationale:
  - In a large diversified portfolio
  - Idiosyncratic risk is diversified away
  - Only systematic risk remains relevant
- Appropriate when:
  - Evaluating funds as components of a broader risky portfolio
  - Selecting managers for inclusion in diversified allocation
- Treynor measures: Return per unit of market risk
- Treynor is appropriate when ranking many pieces that will be mixed to form the overall risky portfolio.

# Information Ratio

- Information Ratio: $$IR = \frac{E(R_p - R_b)}{\sigma(R_p - R_b)}$$
- Where:
  - $R_b$ = benchmark return
  - $R_p - R_b$ = active return ("alpha relative to benchmark")
  - $\sigma(R_p - R_b)$ = tracking error (active risk)
- Active return per unit of benchmark-relative risk
- Sharpe ratio of the active component
- Information Ratio is appropriate when:
  - Investor already holds a diversified benchmark portfolio
  - Portfolio $p$ is evaluated as an active overlay
  - The decision is whether the manager improves the benchmark allocation

# Sharpe Improvement from Active Management

- If benchmark Sharpe is $S_b$, then:

$$S_{new}^2 = S_b^2 + IR^2$$

- Implication:
  - Active strategies improve investor welfare only through positive IR
- Conclusion:
  - $IR$ determines the value of active management around an indexed core.

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# Jensen's Alpha

- CAPM benchmark: $\alpha_p = E(R_p) - [R_f + \beta_p(E(R_M) - R_f)]$

- Alpha measures:
  - Abnormal return beyond systematic exposure
  - Outperformance relative to passive benchmark
- If: $\alpha_p = 0$
  - Then portfolio earns only fair compensation for risk.
- To outperform passive investing:
  - Positive, statistically significant alpha is required
- Alpha is model-dependent and subject to estimation error.

# Performance Ratio Example

- Assume:
  - $R_b$=$R_M$, $R_f$=2%, E[$R_M$]=8%, $\sigma_M$=15%
- Two portfolios:
  - Portfolio A:
    - $\beta_A$=1.1, $\alpha_A$=2.5%, $\sigma(e_A)$=8%
    - $E[R_A]$=11.1%, $\sigma_A$=18.3%
    - Active return relative to benchmark: $E[R_A - R_M]$=3.3%
  - Portfolio B:
    - $\beta_B$=1.0, $\alpha_B$=1.2%,$\sigma(e_B)$=2%
    - $E[R_B]$=9.2%,$\sigma_B$=15.1%
    - Active return relative to benchmark: $E[R_B - R_M]$=1.2%

# Example: Sharpe Ratio Comparison

- Sharpe Ratio
  - Formula: $S_p = \frac{E(R_p - R_f)}{\sigma_p}$
  - Portfolio A: $S_A = \frac{11.1 - 2}{18.3} = \frac{9.1}{18.3} = 0.50$
  - Portfolio B: $S_B = \frac{9.2 - 2}{15.1} = \frac{7.2}{15.1} = 0.48$
- So, Portfolio A preferred as the overall risky portfolio.

# Example: Treynor Measure Comparison

- Treynor Measure
  - Formula: $$T_p = \frac{E(R_p - R_f)}{\beta_p}$$
  - Portfolio A: $$T_A = \frac{11.1 - 2}{1.1} = \frac{9.1}{1.1} = 8.27$$
  - Portfolio B: $$T_B = \frac{9.2 - 2}{1.0} = \frac{7.2}{1.0} = 7.20$$
- So, Portfolio A is ranked as a better portfolio to add into an overall risky portfolio

# Example: Information Ratio Comparison

- Information Ratio
  - Formula: $$IR = \frac{E(R_p - R_b)}{\sigma(R_p - R_b)}$$
  - Portfolio A: $$IR_A = \frac{3.3}{8.14} = 0.41$$
  - Portfolio B: $$IR_B = \frac{1.2}{2.0} = 0.60$$
- However, Portfolio B would be better to mix with the diversified benchmark portfolio

# Connection Between Performance Ratios and Alpha

- Start from the single-index model:
  - $r_p = \alpha_p + \beta_p r_M + e_p$
  - with: $E[e_p]=0$, $Cov(e_p, r_M)=0$
- Treynor Ratio
  - Definition: $T_p = \frac{E(r_p) - r_f}{\beta_p}$
  - Substitute CAPM decomposition:

$$\frac{E(r_p) - r_f}{\beta_p} = \frac{\alpha_p}{\beta_p} + \frac{\beta_p E(r_M - r_f)}{\beta_p}$$

$$T_p = \frac{\alpha_p}{\beta_p} + T_M$$

  - Improvement relative to market: $T_p - T_M = \frac{\alpha_p}{\beta_p}$
- Treynor exceeds the market if and only if $\alpha_p > 0$

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# Connection Between Performance Ratios and Alpha (con.)

- Sharpe Ratio
  - Definition: $$S_p = \frac{E(r_p) - r_f}{\sigma_p}$$
  - Using decomposition: $$E(r_p - r_f) = \alpha_p + \beta_p E(r_M - r_f)$$ $$S_p = \frac{\alpha_p}{\sigma_p} + \rho_{pM} S_M$$
  - Where: $$\rho_{pM} = \mathrm{Corr}(R_p, R_M)$$
  - Improvement relative to market: $$S_p - S_M = \frac{\alpha_p}{\sigma_p} - (1 - \rho_{pM}) S_M$$
- Sharpe improves only if alpha is sufficiently large relative to total volatility.

# Connection Between Performance Ratios and Alpha (con.)

- Information Ratio
  - Definition: $IR = \frac{E(R_p - R_M)}{\sigma(R_p - R_M)}$
  - From single-index model: $R_p - R_M = \alpha_p + e_p$
  - Thus: $IR = \frac{\alpha_p}{\sigma(e_p)}$
  - Improvement over benchmark depends only on alpha relative to residual risk

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# Performance Ratios and Alpha

- All performance measures ultimately depend on $\alpha_p$
- But they scale alpha by different risk measures:
  - Treynor: scales by systematic risk ($\beta p$)
  - Sharpe: scales by total risk ($\sigma_p$)
  - Information Ratio: scales by residual risk ($\sigma(e_p)$)

- Different ratios answer different economic questions about the same alpha.

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# Realized Returns vs. Expected Returns

- When evaluating performance:
  - We do not observe the manager's true expectations
  - We only observe realized returns
  - Returns contain substantial statistical noise
- Observed alpha: $\hat{\alpha} = \alpha + \text{sampling error}$
- Even if true alpha is positive: $\alpha$>0
- Sampling variation may obscure skill.
- To avoid false conclusions, we test: $H_0$: $\alpha$=0
- Skill must be statistically distinguishable from luck.

# Standard Error of Alpha

- Security Characteristic Line (SCL): $r_{p,t}=\alpha+\beta r_{M,t}+e_t$
- Assume: $E[e_t]=0$, $Var(e_t)=\sigma^2(e)$
- Standard error of alpha estimate: $$SE(\hat{\alpha}) = \frac{\sigma(e)}{\sqrt{N}}$$
  - where:
    - $N$ = number of observations
    - $\sigma(e)$ = residual (nonsystematic) volatility
  - t-statistic: $$t(\hat{\alpha}) = \frac{\hat{\alpha}}{SE(\hat{\alpha})} = \frac{\hat{\alpha}\sqrt{N}}{\sigma(e)}$$
- So, statistical power increases only with $\sqrt{N}$

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# How Long to Detect Skill?

- Suppose: $\hat{\alpha} = 0.2\%$ per month
  - 2.4% per year before compounding!
- Residual volatility: σ(e)=2% per month
- Require 5% significance, so *t*=1.96
- Solve:

$$1.96 = \frac{0.2\sqrt{N}}{2}$$

$$N = 384 \text{ months}$$

$$= 32 \text{ years}$$

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# Skill Is Hard to Prove

- Even with:
  - Constant true alpha
  - Stable beta
  - No structural breaks
  - Perfect regression specification
- It take over three decades in the example to statistically confirm skill!
- In reality:
  - Parameters change over time
  - Markets are nonstationary
  - Managers change strategies
  - Sample periods rarely “behave perfectly”
- Conclusion is that moderate statistical noise makes performance evaluation extremely difficult.
- Typical manager tenure < 5 years
- Most careers end before statistical proof of skill is possible.

# Selection Bias and Portfolio Evaluation

- A further complication is that Performance evaluation suffers from selection bias.
- We do not observe all managers — only a selected subset.
- Survivorship Bias
  - Poorly performing funds:
    - Close
    - Merge
    - Disappear from databases
  - Observed sample over-represents successful managers.
  - Average performance is overstated.

# Selection Bias and Portfolio Evaluation (con.)

- Backfill Bias (Incubation Bias)
  - New funds:
    - Often report only after strong early performance
    - Poor early track records are unreported
  - Database history is selectively constructed.
- Self-Selection Bias
  - Managers with:
    - Strong returns advertise performance
    - Weak returns remain silent
  - Published performance not random sample.
- Data-Snooping and Multiple Testing
  - With thousands of managers:
  - Some will appear to have high alpha by chance
  - If: $\alpha$=0 for all managers, a fraction will still show $t$>2 purely due to randomness.

# Style Analysis: What Is the Manager Actually Doing?

- Before attributing performance to skill, ask: Is outperformance due to security selection or asset allocation?
- Asset Class Style Exposure:
  - A portfolio's return may largely reflect exposure to:
    - U.S. large-cap equities
    - Small-cap equities
    - Growth vs. value stocks
    - International equities
    - Emerging markets
    - Fixed income duration
    - Credit risk
    - Cash holdings
  - Apparent "alpha" may simply reflect being overweight a winning asset class.

# Style Analysis: What Is the Manager Actually Doing? (con.)

- Manager claims stock-picking skill.
- But analysis shows:
  - Persistent tilt toward small-cap value
  - Small-cap value outperforms during evaluation period
- Result: $\hat{\alpha} > 0$
- But true driver is systematic exposure, not security selection

# Style Analysis: What Is the Manager Actually Doing? (con.)

- Returns-Based Style Analysis
  - Purpose is to separate asset allocation (style) from security selection skill
  - Regression Framework:
    - Regress fund returns on a set of asset class index returns: $$R_{p,t} = \alpha + \sum_{k=1}^{K} w_k R_{k,t} + \epsilon_t$$
      - where:
        - $R_{k,t}$ = returns on asset class indexes
        - $w_k$ = implicit exposure to asset class $k$
  - Interpretation of Coefficients (Constraint: long-only mutual funds): $$w_k \geq 0, \quad \sum_{k=1}^{K} w_k = 1$$
    - Estimated $w_k$ ≈ fund's effective asset allocation
    - Regression recovers "style weights"
    - Reveals whether fund behaves like its stated mandate
- Style = asset allocation pattern inferred from returns.

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# Style Analysis: What Is the Manager Actually Doing? (con.)

- Interpreting $R^2$
  - $R^2$=fraction of return variability explained by style
  - High $R^2$
    - Performance largely driven by asset allocation
    - Little room for security selection impact
  - Low $R^2$
    - Greater unexplained variation
    - May reflect:
      - Security selection within asset classes
      - Market timing
      - Style drift over time

# Style Analysis: What Is the Manager Actually Doing? (con.)

- Intercept = $\alpha$
- Measures average return beyond style exposures.
- However:
  - If style factors fully capture systematic risk, alpha reflects abnormal performance.
  - If relevant risk factors omitted, alpha may be mismeasured.
- Observed outperformance may reflect being in the right asset classes rather than superior security selection skill
- Style analysis helps determine whether “alpha” is truly abnormal or simply compensation for systematic asset allocation choices.

# Changing Portfolio Strategy During the Evaluation Period

- Performance measures typically assume:
  - Stable risk exposure
  - Stable investment style
  - Constant beta
- In reality:
  - beta is not constant
  - some manager change strategies over time

# Changing Portfolio Strategy During the Evaluation Period

- Example; Risk-shifting
  - Suppose a manager:
    - Uses conservative strategy in downturns
    - Shifts to aggressive strategy when expecting strong markets
      - Subperiod 1: $\mu_1$, $\sigma_1$
      - Subperiod 2: $\mu_2$, $\sigma_2$
      - Each subperiod may have high Sharpe:

$$S_1 = \frac{\mu_1 - r_f}{\sigma_1}$$

$$S_2 = \frac{\mu_2 - r_f}{\sigma_2}$$

      - Over the full sample:

$$\bar{\mu} = \frac{\mu_1 + \mu_2}{2}$$

$$\bar{\sigma} > \frac{\sigma_1 + \sigma_2}{2}$$

      - Thus $S_{overall} = \frac{\bar{\mu} - r_f}{\bar{\sigma}}$ may be lower than either subperiod Sharpe.

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# Changing Portfolio Strategy During the Evaluation Period

- Why This Happens:
  - Volatility is nonlinear
  - Combining regimes increases variance
  - Strategy switching can distort full-period Sharpe
  - Sharpe assumes stationary return distribution
  - Violation of stationarity biases evaluation.
- Changing exposure can:
  - Understate true skill
  - Overstate market timing
  - Distort alpha estimates
- Performance evaluation is sensitive to time variation in risk exposures
- Stable strategy assumption is critical in performance measurement.

# Performance Manipulation and Return Smoothing

- Managers are evaluated on:
  - Sharpe ratios
  - Alpha
  - Downside risk
  - Peer rankings
- Incentives may encourage manipulation of reported risk
- Common Forms of Manipulation:
  - Return Smoothing
    - Illiquid assets marked infrequently
    - Use of models rather than market prices
    - Stale pricing
  - Effect: $\sigma_{reported} < \sigma_{true}$ - thus Sharpe ratio artificially inflated.

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# Performance Manipulation and Return Smoothing (con.)

- Tail Risk Strategies
  - Selling out-of-the-money options
  - Carry trades
  - Leveraged credit exposure
  - Produces:
    - Frequent small gains
    - Occasional large losses
    - Sharpe appears high — until crash occurs.
- Window Dressing
  - Adjust portfolio holdings before reporting date
  - Sell poor performers
  - Buy recent winners
  - Reported holdings may not reflect actual strategy

# Morningstar Risk-Adjusted Rating (MRAR)

- Morningstar Star Ratings are based on a risk-adjusted performance measure:

$$MRAR = \left[ \frac{1}{T} \sum_{t=1}^{T} \left( \frac{1 + R_{p,t}}{1 + R_{f,t}} \right)^{-\gamma} \right]^{-\frac{1}{\gamma}} - 1$$

  - Where:
  - $R_{p,t}$ = portfolio return in period $t$
  - $R_{f,t}$ = risk-free rate
  - $\gamma$ = risk-aversion parameter
  - $T$ = number of observations
- Note that Morningstar uses $\gamma$ = 2 as a reasonable value for an average retail client

# Morningstar Risk-Adjusted Rating (MRAR) (con.)

- Utility-based performance measure
- Penalizes downside volatility more heavily
- Incorporates investor risk aversion
- Why It Is Widely Used
  - Accounts for volatility and downside risk
  - Comparable across funds within peer group
  - Forms basis for Morningstar “star ratings”
  - Among the most widely used performance measures for mutual funds

# Market Timing: Value and Difficulty

- Market timing attempts to adjust exposure to the market based on forecasts of:
  - Expected market return
  - Volatility
  - Business cycle conditions
- Instead of constant beta:

$$r_{p,t} = \alpha + \beta_t r_{M,t} + e_t$$

- Timing implies:

$$\beta_t \text{ increases when } E[r_{M,t}] \text{ is high}$$

$$\beta_t \text{ decreases when } E[r_{M,t}] \text{ is low}$$

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# Measuring Market Timing

- A common test augments the regression:

$$r_{p,t} = \alpha + \beta r_{M,t} + \gamma r_{M,t}^2 + e_t$$

- So, if $\gamma$>0
  - convex payoff
  - Portfolio increases exposure in up markets
  - Evidence of successful timing
- But, if $\gamma$=0
  - No timing ability detected

# Why Timing Is Difficult

- Timing requires:
  - Accurate forecasts of expected returns
  - Accurate forecasts of volatility
  - Rapid adjustment of portfolio weights
  - Low transaction costs
- Empirical evidence:
  - Most managers show little consistent timing ability
  - Positive timing coefficients rare and unstable
  - After costs, net benefit often negligible
- If timing signals were easily exploitable:
  - Prices would adjust
  - Expected abnormal returns would disappear

# Performance Attribution: Where Did Returns Come From?

- Excess return relative to benchmark = $R_p - R_b$
  - Let:
    - $w_{p,i}$ = portfolio weight in asset class $i$
    - $w_{b,i}$ = benchmark weight in asset class $i$
    - $R_{p,i}$ = portfolio return in asset class $i$
    - $R_{b,i}$ = benchmark return in asset class $i$
- Asset Allocation Effect:

$$\sum_{i=1}^{n}(w_{p,i} - w_{b,i})R_{b,i}$$

- Measures value added from:
  - Overweighting or underweighting asset classes
  - Style tilts relative to benchmark
  - Example:
    - Overweight equities vs bonds
    - Overweight emerging markets

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# Performance Attribution: Where Did Returns Come From? (con.)

- Security Selection Effect

$$\sum_{i=1}^{n} w_{b,i}(R_{p,i} - R_{b,i})$$

- Measures value added from:
  - Stock picking within asset classes
  - Manager skill relative to benchmark constituents
  - Example: Choosing superior firms within U.S. large-cap sector
- Total Excess Return
  - $R_p - R_b$=Asset Allocation Effect + Security Selection Effect
- Attribution separates systematic allocation decisions from security-level skill

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# Performance Attribution Example

| Asset Class | Benchmark Weight ($w_b$) | Portfolio Weight ($w_p$) | Benchmark Return ($R_b$) | Portfolio Return ($R_p$) |
|---|---|---|---|---|
| Equities | 0.60 | 0.70 | 8.00% | 9.25% |
| Bonds | 0.30 | 0.25 | 4.00% | 4.20% |
| Cash | 0.10 | 0.05 | 2.00% | 2.00% |

- Benchmark total return:
  - $R_b$=0.60(8.00)+0.30(4.00)+0.10(2.00)=6.20%
- Portfolio total return:
  - $R_p$=0.70(9.25)+0.25(4.20)+0.05(2.00)=7.625%
- Total excess return:
  - $R_p - R_b$=7.625%−6.20%=1.425%=142.5 bp

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# Performance Attribution Example (con.)

- Asset Allocation Effect
  - Equities: (0.70−0.60)(8.00)=0.10(8.00)=0.80%=80 bp
  - Bonds: (0.25−0.30)(4.00)=−0.05(4.00)=−0.20%=−20 bp
  - Cash: (0.05−0.10)(2.00)=−0.05(2.00)=−0.10%=−10 bp
  - Total allocation:80−20−10=50 bp80−20−10=50 bp
- Security Selection Effect
  - Equities: 0.70(9.25−8.00)=0.70(1.25)=0.875%=87.5 bp
  - Bonds: 0.25(4.20−4.00)=0.25(0.20)=0.05%=5.0 bp
  - Cash: 0.05(2.00−2.00)=0
  - Total selection: 87.5+5.0=92.5 bp
- Total excess return: 50 bp + 92.5 bp = 142.5 bp